

The sum $\sum_{j=i_0+1}^{i_1} \mathbb{1}\{j \in \mathcal{B}_C\}$ is thus stochastically dominated by a sum of independent $\{0, 1\}$ -random variables

$$Z = \sum_{i=0}^{i_{\max}-1} \sum_{h=1}^{\omega} Z_h^{(i)},$$

where $\Pr(Z_h^{(i)} = 1) = (1 + o(1)) \left(1 - \frac{2t_i}{3}\right)$ for all $h = 1, 2, \dots, \omega$, uniformly in i . We compute

$$\begin{aligned} \mathbb{E}[Z] &= (1 + o(1))\omega \sum_{i=0}^{i_{\max}-1} \left(1 - \frac{2t_i}{3}\right) \\ &= (1 + o(1))\omega \sum_{i=0}^{i_{\max}-1} \left(1 - \frac{2}{3} \cdot \frac{i_0 + i\omega}{n}\right) \\ &= (1 + o(1))\omega i_{\max} \left(1 - \frac{2i_0}{3n}\right) - \frac{2\omega}{3n} \sum_{i=0}^{i_{\max}-1} i \\ &= (1 + o(1)) \left(n - \frac{2\omega}{3n} \frac{i_{\max}^2}{2}\right) \\ &= (1 + o(1)) \frac{2n}{3}. \end{aligned}$$

It follows from Proposition 1.5 that $Z \leq (1 + o(1))\mathbb{E}[Z] = (1 + o(1))\frac{2n}{3}$ a.a.s. Since Z stochastically dominates $|\mathcal{B}_C|$, we may conclude $\frac{1}{2}|\mathcal{B}_C| \leq (1 + o(1))\frac{n}{3}$ a.a.s., completing the proof. $\square$

7. Conclusion

In this paper, we have shown that $s(\mathcal{G}_{n,3}) \leq (1 + o(1))\frac{n}{3}$ a.a.s., and have additionally made the conjecture that $s(\mathcal{G}_{n,3}) = (1 + o(1))\frac{n}{4}$ a.a.s., which is (asymptotically) smallest possible for cubic graphs on n vertices. Resolving this conjecture is an interesting direction for future research. Another one is to study $s(\mathcal{G}_{n,d})$ for $d \geq 4$ —a problem which at present appears challenging.

We considered the following crude technique for upper bounding the Sudoku number of a graph G with chromatic number k . Start with any k -coloring of G using colour classes $\{1, 2, \dots, k\}$. Iteratively move vertices from class k to another available class until every remaining vertex has a neighbour in each class i for $i = 1, \dots, k-1$. Then the union of classes 1 through $k-1$ is a Sudoku set for G , and we have the bound

$$s(G) \leq (\chi(G) - 1)\alpha(G), \tag{38}$$

where $\alpha(G)$ is the size of the largest independent set in G .

Using the best available bound $\alpha(\mathcal{G}_{n,4}) \leq 0.41635$ a.a.s. (see [23]) and the fact that $\chi(\mathcal{G}_{n,4}) = 3$ a.a.s. ([26]), (38) gives $s(\mathcal{G}_{n,4}) \leq 0.8327n$ a.a.s. The next value of d for which $\chi(\mathcal{G}_{n,d})$ is known explicitly is $d = 6$, where we have $\chi(\mathcal{G}_{n,d}) = 4$ a.a.s. [27]. However, it has been shown that $\mathcal{G}_{n,6}$ has an independent set of size at least $0.33296n$ a.a.s. [8], meaning that any bound using (38) for $d = 6$ will be very close to n . In fact, currently available upper bounds, such as $0.35799n$, are well above $n/3$ (see for instance [8]) and thus, are far from providing any useful bound on $s(\mathcal{G}_{n,6})$. For large d , (38) is too weak to say anything nontrivial about $s(\mathcal{G}_{n,d})$. Indeed, for $d \rightarrow \infty$ sufficiently slowly with respect to n , it is known that $\chi(\mathcal{G}_{n,d}) = (1 + o(1))\frac{n}{\alpha(\mathcal{G}_{n,d})} = (1 + o(1))\frac{d}{2\log d}$ a.a.s. [10].